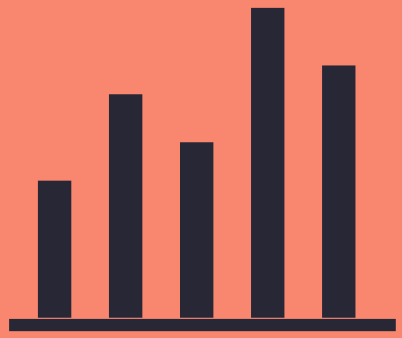
PYTHON FOR DATA SCIENCE · PROJECT SHOWCASE

AUTOMATED TECH SALARY & SKILLS ANALYZER



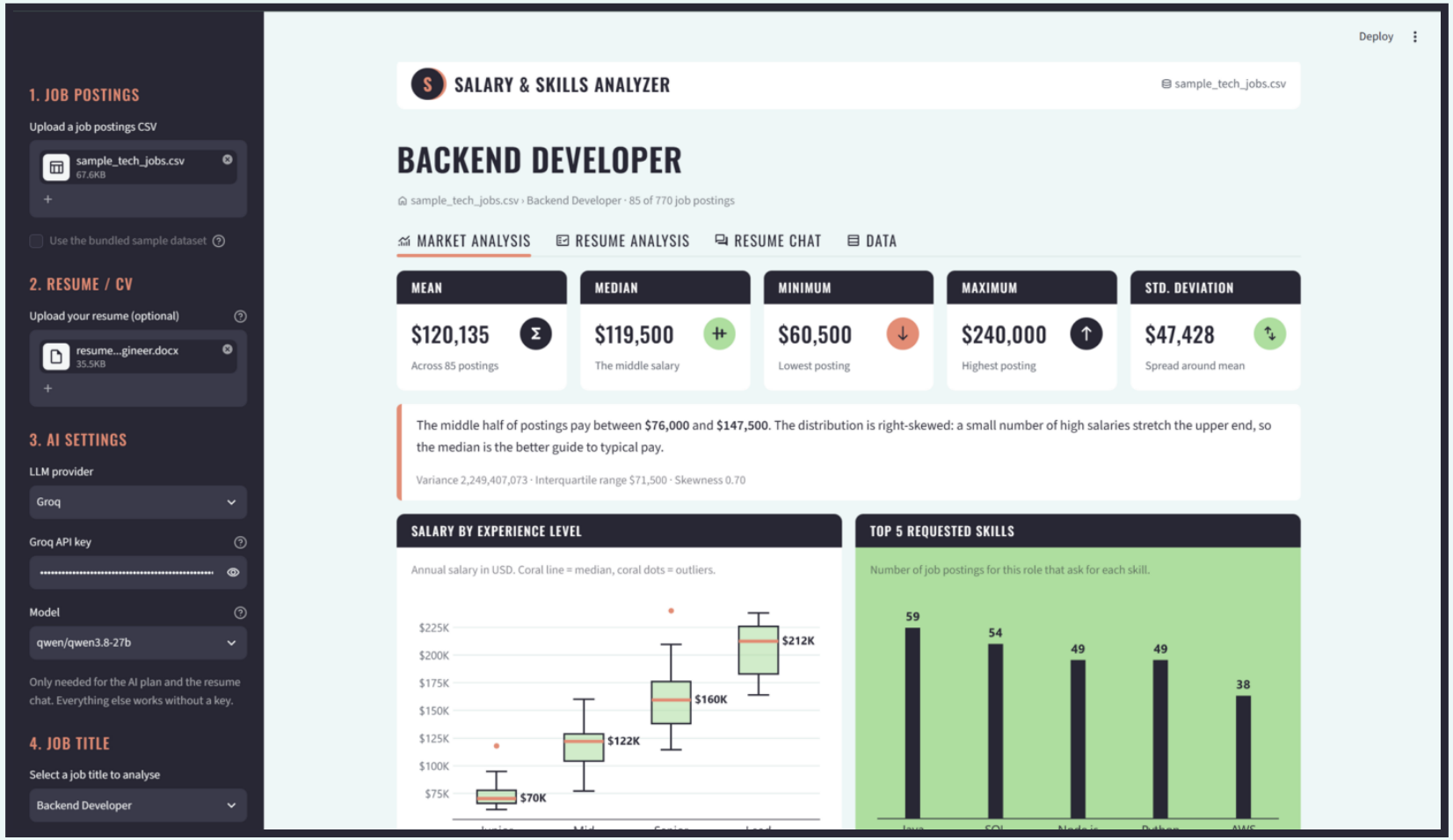
A Streamlit web app that turns tech job postings into salary statistics and skill demand, checks your resume against the market, and lets you ask an AI career coach what to improve.

Streamlit | Pandas | NumPy | SciPy | Matplotlib | Seaborn | LangChain

Siddharth Devmurari

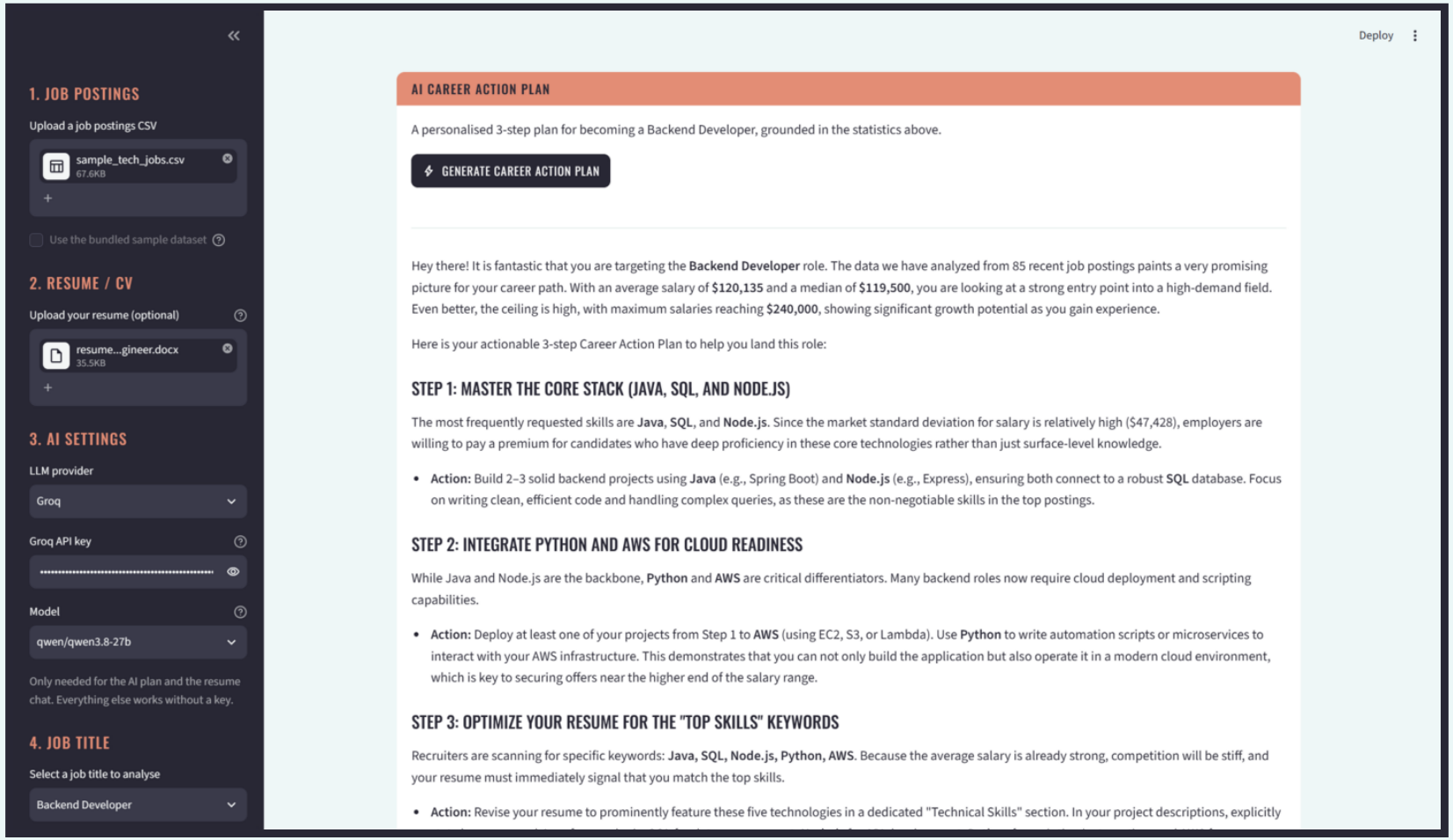
github.com/SiddharthDevmurari/Automated-Tech-Salary-Skills-Analyzer

MARKET ANALYSIS



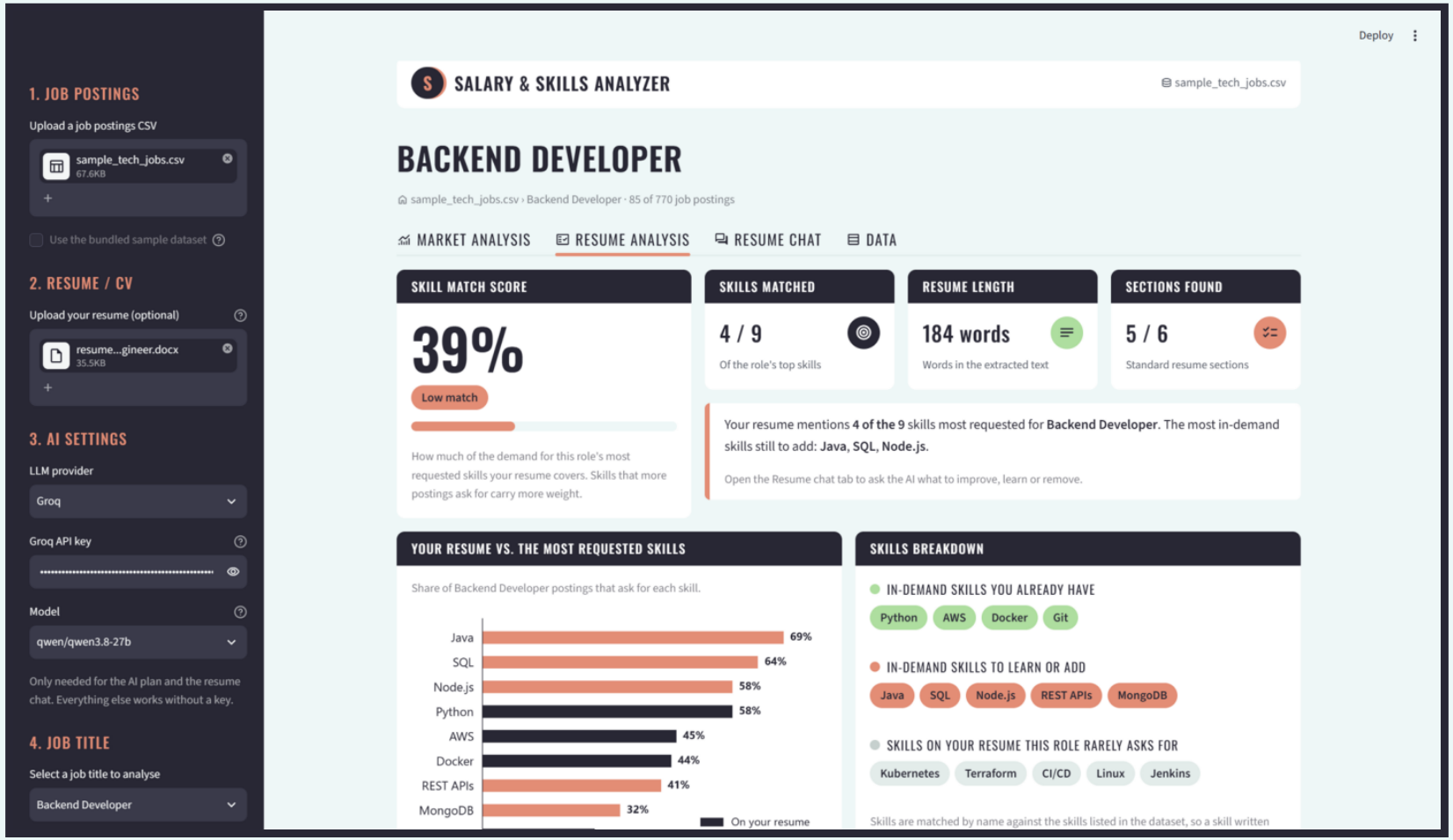
Pick a job title and the app computes mean, median, minimum, maximum and standard deviation of salaries with Pandas and SciPy, then charts the pay spread by experience level (Seaborn boxplot) and the five most requested skills (Matplotlib).

AI CAREER ACTION PLAN



The computed statistics are passed through a LangChain chain to an LLM, which writes a 3-step plan that quotes the actual numbers: average and maximum salary, salary spread and the top skills for the role.

RESUME ANALYSIS: SKILL MATCH SCORE



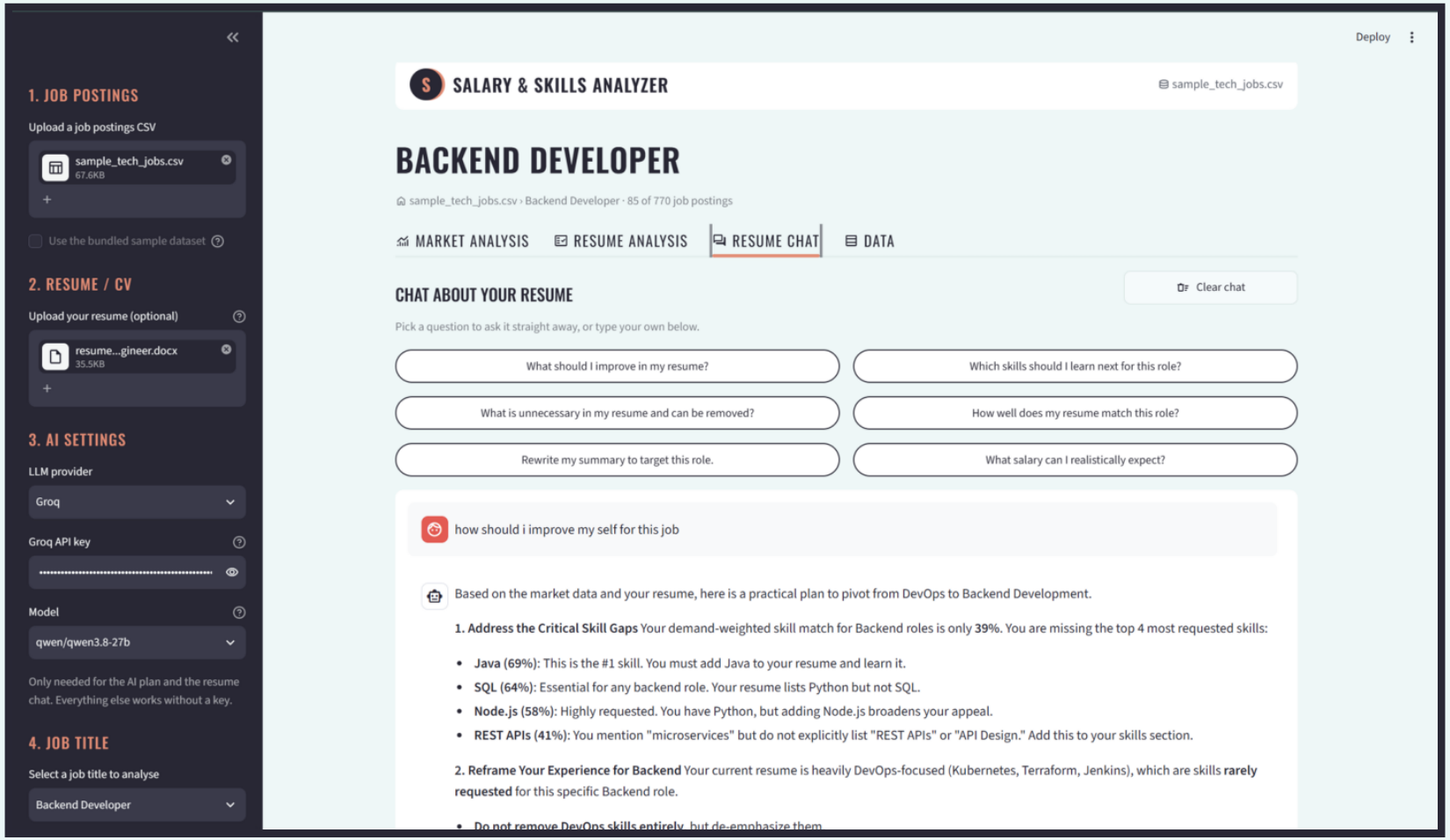
Upload a resume (PDF, DOCX or TXT). The app finds which skills it mentions and scores it against the role's most requested skills, weighted by how many job postings ask for each one.

RESUME ANALYSIS: SKILL GAPS AND CHECKLIST



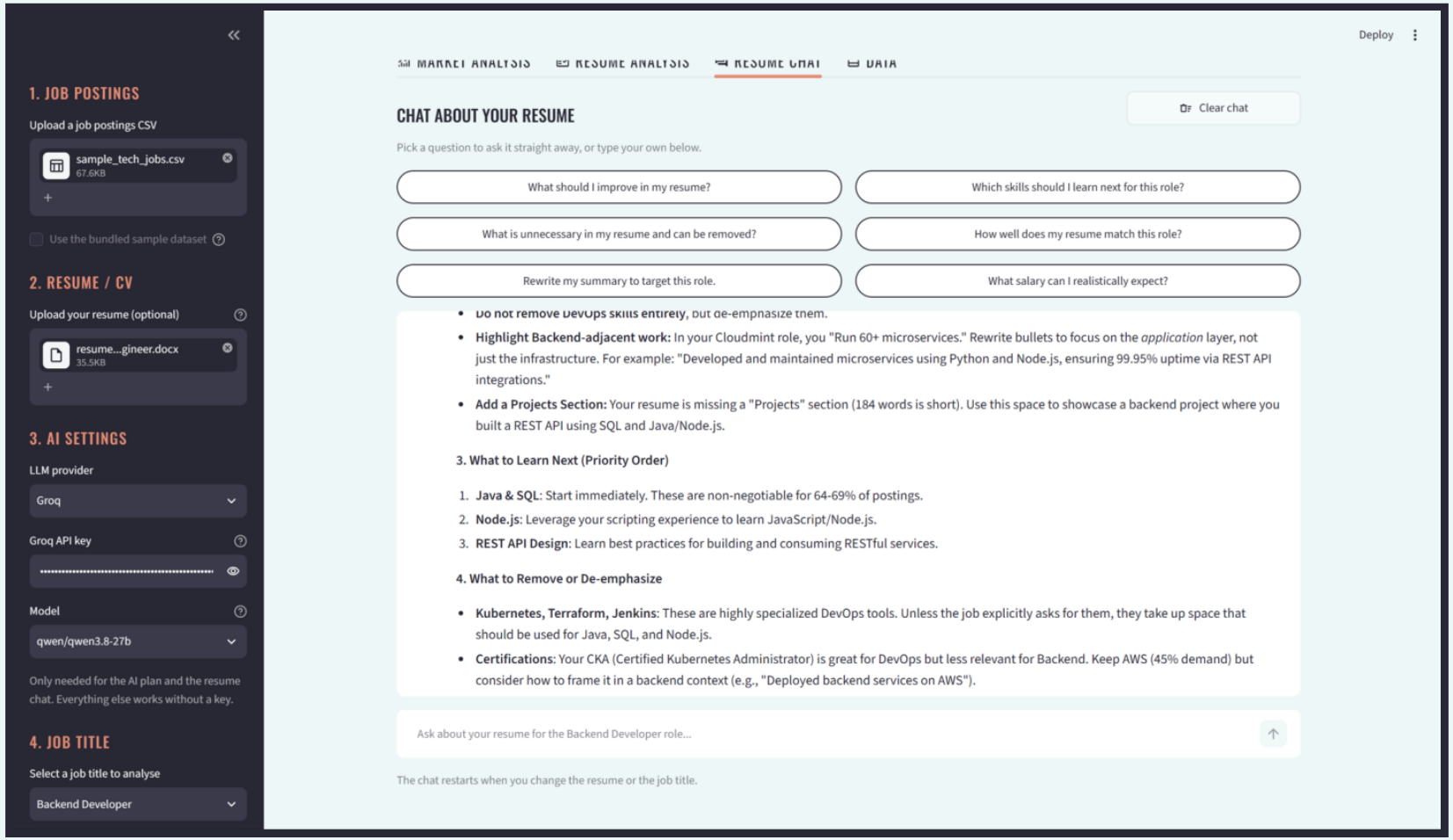
Skills you already have, skills to learn, and skills this role rarely asks for, shown beside the demand for each.
A checklist flags missing resume sections and contact links.

RESUME CHAT WITH ONE-CLICK QUESTIONS



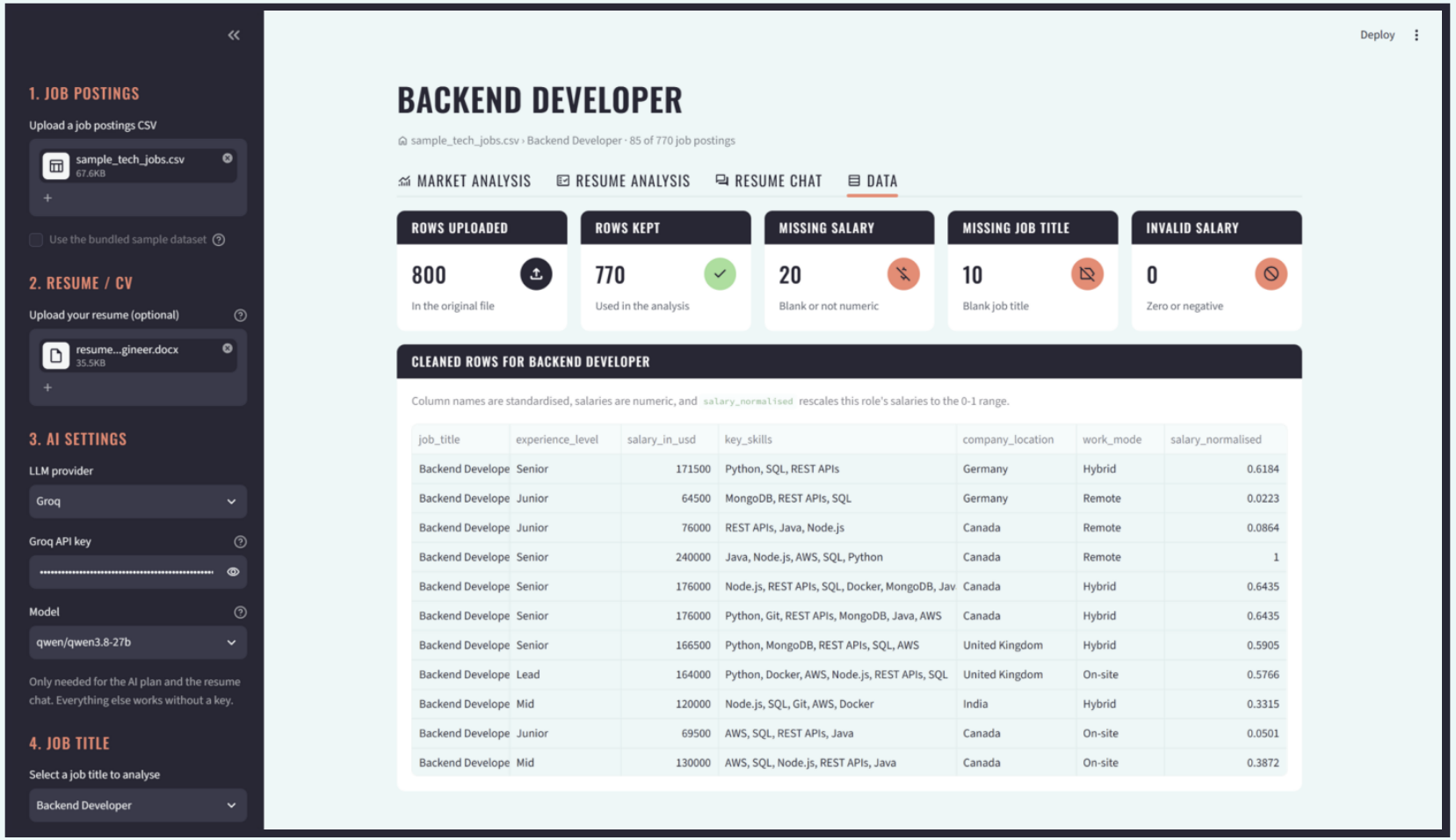
Chat with the AI about your resume. Sample questions send in one click, and every answer is grounded in the resume text and the market data for the selected role.

RESUME CHAT: WHAT TO LEARN AND WHAT TO REMOVE



The AI prioritises what to learn next and points out what is unnecessary for the target role, referring to specific lines of the resume.

DATA CLEANING



Before any analysis, rows with a missing or invalid salary or job title are dropped and salaries are normalised.
The Data tab shows what was removed and the cleaned rows.

HOW IT WORKS

1. UPLOAD

A CSV of job postings and, optionally, your resume (PDF, DOCX or TXT).

2. CLEAN

Pandas fixes column names, converts salaries and drops missing or invalid rows.

3. ANALYSE

Pandas, NumPy and SciPy compute salary statistics and skill demand.

4. VISUALISE

Seaborn and Matplotlib draw the boxplot and the skill charts.

5. ADVISE

LangChain sends the results to an LLM for a career plan and the resume chat.

WHAT IT DEMONSTRATES

- Data preparation with Pandas: missing values, type conversion, filtering and normalisation.
- Descriptive statistics with Pandas, NumPy and SciPy: central tendency, dispersion, quartiles and skewness.
- Data visualisation with Matplotlib and Seaborn, embedded in a Streamlit dashboard.
- LLM integration with LangChain (LCEL): prompt templates, streaming chat and answers grounded in computed data.

Source code: github.com/SiddharthDevmurari/Automated-Tech-Salary-Skills-Analyzer

Screenshots use the app's bundled sample dataset, which is synthetic.